

Math 135

Please give me feedback! I am at the first row.

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4 Proving Mathematical Statement

4.1 Proving Universally Quantified Statements

A **proposition** is a true statement that has been proved to be true. A **theorem** is a particularly significant proposition. A **lemma** is a subsidiary proposition. A **corollary** is a proposition that follows almost immediately from a theorem.

Example 4.1.1

The law of cosines is a theorem: in a triangle with sides a, b, c , where θ is the angle between a and b ,

$$c^2 = a^2 + b^2 - 2ab \cos \theta.$$

The Pythagorean theorem is a corollary: taking $\theta = \pi/2$ gives $\cos \theta = 0$, so $c^2 = a^2 + b^2$.

To prove $\forall x \in S, P(x)$, let x be an *arbitrary* element of S , using only the fact that $x \in S$, and show that $P(x)$ is true. Since nothing special about x was assumed, the argument works for every element of S .

Let's do our first proof.

Proposition 4.1.2

$$\forall x \in \mathbb{R}, x^2 + 1 \geq 2x.$$

Remark (Proof idea)

Moving everything to one side gives $x^2 - 2x + 1 \geq 0$, and the left side is the perfect square $(x - 1)^2$. Since squares of real numbers are never negative, we start from $(x - 1)^2 \geq 0$ and work forward.

Proof

Let x be an arbitrary real number. Therefore, $x - 1$ must also be a real number, and hence

$$(x - 1)^2 \geq 0.$$

Expanding the left side, we get

$$x^2 - 2x + 1 \geq 0.$$

Adding $2x$ to both sides yields

$$x^2 + 1 \geq 2x.$$

Q.E.D.

Proposition 4.1.3

$$\forall x, y \in \mathbb{R}, x^4 + x^2y + y^2 \geq 5x^2y - 3y^2.$$

Remark (Proof idea)

Moving everything to the left side gives $x^4 - 4x^2y + 4y^2 \geq 0$, and $x^4 - 4x^2y + 4y^2 = (x^2 - 2y)^2$. This explains the choice of $x^2 - 2y$ below.

Proof

Let x and y be arbitrary real numbers. Then $x^2 - 2y$ is also a real number. So

$$(x^2 - 2y)^2 \geq 0.$$

Expanding, we get

$$x^4 - 4x^2y + 4y^2 \geq 0.$$

Adding $5x^2y$ to both sides, we have

$$x^4 + x^2y + 4y^2 \geq 5x^2y.$$

Now, subtract $3y^2$ from both sides:

$$x^4 + x^2y + y^2 \geq 5x^2y - 3y^2.$$

Q.E.D.

Proposition 4.1.4

$$\forall x, y \in \mathbb{R}, \max(x, y) = \frac{x + y + |x - y|}{2}.$$

Remark (Proof idea)

Both $\max(x, y)$ and $|x - y|$ are defined by cases depending on which of x, y is larger, so we split into the cases $x \geq y$ and $x < y$. These cases cover every pair of real numbers.

Proof

Let x and y be arbitrary real numbers.

Case 1: If $x \geq y$, then $\max(x, y) = x$, and $x - y \geq 0$, so $|x - y| = x - y$.

Then

$$\frac{x + y + |x - y|}{2} = \frac{x + y + x - y}{2} = \frac{2x}{2} = x = \max(x, y).$$

So the claim holds when $x \geq y$.

Case 2: If $x < y$, then $\max(x, y) = y$, and $x - y < 0$. So

$$|x - y| = -(x - y) = y - x.$$

Then

$$\frac{x + y + |x - y|}{2} = \frac{x + y + y - x}{2} = \frac{2y}{2} = y = \max(x, y).$$

The claim holds in both cases, so it holds for all $x, y \in \mathbb{R}$.

Q.E.D.

4.1.1 Disproving a Universal Statement

Since $\neg(\forall x \in S, P(x)) \equiv \exists x \in S, \neg P(x)$, to disprove a universal statement it suffices to give one **counterexample**: a specific $x \in S$ for which $P(x)$ is false.

Example 4.1.5

Disprove:

$$\forall x \in \mathbb{R}, (x^2 - 1)^2 > 0.$$

Solution. Take $x = 1$. Then

$$(x^2 - 1)^2 = (1 - 1)^2 = 0 \not> 0.$$

So $x = 1$ is a counterexample, and the statement is false.

Exercises

- Prove $\forall \theta \in \mathbb{R}, \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$.
- Prove $\forall x \in \mathbb{R}, |x - 3| + 2|x + 2| \geq 5$.

4.2 Proof by Contrapositive, Elimination

Several examples below use divisibility.

Definition 4.2.1 (Divisibility)

Let $a, b \in \mathbb{Z}$. We say a **divides** b , written $a \mid b$, if

$$\exists k \in \mathbb{Z}, b = ak.$$

Otherwise we write $a \nmid b$.

Definition 4.2.2 (Proof by Contrapositive)

Since $(A \implies B) \equiv (\neg B \implies \neg A)$, we can prove $(A \implies B)$ by proving that $((\neg B) \implies (\neg A))$. This method is useful when a direct proof of $A \implies B$ is awkward, but assuming $\neg B$ gives a concrete hypothesis that is easier to work with.

Example 4.2.3

Here is an example of contrapositive proof.

Proposition 4.2.4

For every integer x , if $4 \nmid x$ then $36 \nmid 3x$.

Proof

The hypothesis $4 \nmid x$ says only what x is *not*, which is hard to use directly, so we prove the contrapositive

$$(36 \mid 3x) \implies (4 \mid x).$$

Let x be an arbitrary integer. Assume that $36 \mid 3x$. Thus, there exists an integer k such that $3x = 36k$.

Dividing both sides by 3,

$$x = 12k = 4 \cdot (3k) \text{ and } 3k \in \mathbb{Z},$$

so $4 \mid x$.

Therefore, by the contrapositive method,

$$(\forall x \in \mathbb{Z}) [4 \nmid x \implies 36 \nmid 3x].$$

Q.E.D.

Example 4.2.5

Here is another example.

Proposition 4.2.6

$$(\forall a, b \in \mathbb{R}) [(ab \in \mathbb{R} \setminus \mathbb{Q}) \implies (a \in \mathbb{R} \setminus \mathbb{Q} \vee b \in \mathbb{R} \setminus \mathbb{Q})]$$

Proof

Irrationality is a “not” condition, while rationality gives us fractions to compute with. By De Morgan’s Law, the negation of the conclusion is “ $a \in \mathbb{Q}$ and $b \in \mathbb{Q}$ ”, so the contrapositive is: $\forall a, b \in \mathbb{R}$, if $a \in \mathbb{Q}$ and $b \in \mathbb{Q}$, then $ab \in \mathbb{Q}$.

Let a, b be arbitrary in $\mathbb{R}$. Assume that a, b are both rational.

Then there exist $p, q, r, s \in \mathbb{Z}$ with $q \neq 0$ and $s \neq 0$ such that

$$a = \frac{p}{q} \quad \text{and} \quad b = \frac{r}{s}.$$

Thus,

$$a \cdot b = \frac{pr}{qs} \quad \text{and} \quad qs \neq 0,$$

where $pr, qs \in \mathbb{Z}$. Hence $ab \in \mathbb{Q}$.

By the contrapositive,

$$\forall a, b \in \mathbb{R}, (ab \notin \mathbb{Q}) \implies (a \notin \mathbb{Q} \vee b \notin \mathbb{Q}).$$

Q.E.D.

Definition 4.2.7 (Elimination Method)

Let A, B, C be statement variables.

$$(A \Rightarrow (B \vee C)) \equiv ((A \wedge \neg B) \Rightarrow C)$$

This is useful when the conclusion is a disjunction $B \vee C$: instead of proving the disjunction directly, we assume one disjunct fails ($\neg B$) and show the other (C) must then hold.

Since $B \vee C \equiv C \vee B$, we may instead eliminate C and prove

$$(A \wedge \neg C) \Rightarrow B.$$

Example 4.2.8

Here is an example:

Proposition 4.2.9

Prove that for all $x \in \mathbb{R}$, if $|2x - 6| = 4$, then $x > 3$ or $x = 1$.

Proof

The negation of $x > 3$ is $x \leq 3$, so the elimination form is

$$\forall x \in \mathbb{R}, [(|2x - 6| = 4) \wedge (x \leq 3)] \Rightarrow x = 1.$$

Let x be an arbitrary real number. Assume $|2x - 6| = 4$ and $x \leq 3$, and prove $x = 1$.

Since $|2x - 6| = 4$, thus

$$2x - 6 = \pm 4.$$

So $2x = 10$ or $2x = 2$, meaning

$$x = 5 \quad \text{or} \quad x = 1.$$

Since $x \leq 3$, the value $x = 5$ is ruled out, so

$$x = 1.$$

Q.E.D.

Proposition 4.2.10

$$\forall a, b, c \in \mathbb{Z}, (\forall x \in \mathbb{Z}, a \mid (bx + c)) \Rightarrow (a \mid (c - 2b)).$$

Remark (Proof idea)

The hypothesis holds for *every* integer x , so we may instantiate it at any x we like. Choosing $x = -2$ makes $bx + c$ collapse to $c - 2b$, which is exactly the divisibility claim we want — no other reasoning is needed once $x = -2$ is substituted.

Proof

Let a, b, c be arbitrary integers. Assume for all $x \in \mathbb{Z}$, $a \mid (bx + c)$.

Since this holds for every integer x , it holds in particular when $x = -2$, so we get

$$a \mid (b(-2) + c),$$

that is

$$a \mid (c - 2b).$$

Q.E.D.

Proposition 4.2.11 (Divisibility of Integer Combinations (DIC))

Let $a, b, c \in \mathbb{Z}$. If $a \mid b$ and $a \mid c$, then for all $x, y \in \mathbb{Z}$,

$$a \mid (bx + cy).$$

Remark (Proof idea)

Divisibility means b and c are each an integer multiple of a . Any integer combination $bx + cy$ is then a sum of multiples of a , hence itself a multiple of a — this is exactly what licenses factoring a out below.

Proof

Let $a, b, c \in \mathbb{Z}$. Assume $a \mid b$ and $a \mid c$. Then there exist integers k_1, k_2 such that

$$b = ak_1 \quad \text{and} \quad c = ak_2.$$

Let $x, y \in \mathbb{Z}$ be arbitrary. Then

$$bx + cy = ak_1x + ak_2y = a(k_1x + k_2y).$$

Since $k_1x + k_2y \in \mathbb{Z}$, this shows $a \mid (bx + cy)$.

Q.E.D.

Proposition 4.2.12 (Converse of DIC)

Let $a, b, c \in \mathbb{Z}$. If for all $x, y \in \mathbb{Z}$, $a \mid (bx + cy)$, then $a \mid b$ and $a \mid c$.

Remark (Proof idea)

The hypothesis holds for *every* choice of x, y , so we are free to pick whichever values isolate b and c on their own: $(x, y) = (1, 0)$ leaves $bx + cy = b$, and $(x, y) = (0, 1)$ leaves $bx + cy = c$.

Proof

Let a, b, c be arbitrary integers. Assume that for all $x, y \in \mathbb{Z}$, $a \mid (bx + cy)$.

Taking $x = 1, y = 0$, we get

$$a \mid (b \cdot 1 + c \cdot 0), \quad \text{that is,} \quad a \mid b.$$

Taking $x = 0, y = 1$, we get

$$a \mid (b \cdot 0 + c \cdot 1), \quad \text{that is, } a \mid c.$$

Therefore $a \mid b$ and $a \mid c$.

Q.E.D.

4.3 Proof By Contradiction

Definition 4.3.1

Given a statement A , the logical expression $A \wedge (\neg A)$ is called a contradiction.

A contradiction is false in every row of its truth table. To prove a statement S by contradiction, assume $\neg S$ and derive a contradiction; since a true assumption cannot lead to a false statement, $\neg S$ must be false, so S is true. For an implication $A \implies B$, we assume A together with $\neg B$.

Proposition 4.3.2

$\forall n \in \mathbb{Z}$, if n^2 is even, then n is even.

Proof

Let n be an arbitrary integer.

Assume n^2 is even.

Suppose, for contradiction, that n is odd.

Then,

$$(\exists k \in \mathbb{Z}) n = 2k + 1.$$

It implies

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

but $2k^2 + 2k \in \mathbb{Z}$ (since $k \in \mathbb{Z}$). So we have actually shown that n^2 is odd.

This contradicts the assumption that n^2 is even.

Thus, the assumption that n is odd must be false. Thus, n is even.

Q.E.D.

Exercises

- Prove there is no greatest integer.
- Prove $\forall a, b \in \mathbb{Z}, a \geq 2 \implies (a \nmid b) \vee (a \nmid (b + 1))$.
- Prove $\sqrt{2}$ is irrational.

4.4 Proving Uniqueness

To prove “there is a unique $x \in S$ such that $P(x)$ ”, we prove two things:

- **Existence:** there is at least one $x \in S$ such that $P(x)$ is true (for example, by exhibiting one);

- **Uniqueness:** assume that $P(x)$ and $P(y)$ are both true for some $x, y \in S$, and prove that $x = y$.

Uniqueness alone does not guarantee that such an x exists, so both parts are needed.

Proposition 4.4.1

$\forall a, b \in \mathbb{Z}$, if $a \neq 0$ and $a \mid b$, then there is a unique integer k such that $b = ka$.

Proof

Let a, b be arbitrary in $\mathbb{Z}$. Assume $a \neq 0$ and $a \mid b$.

Existence. Since $a \mid b$, by definition of divisibility,

$$(\exists k \in \mathbb{Z}) b = ka.$$

Uniqueness. Suppose $k, m \in \mathbb{Z}$ satisfy

$$b = ka \quad \text{and} \quad b = ma.$$

Thus, $ka = ma$. Since $a \neq 0$, we may divide both sides by a , and we have

$$k = m.$$

Thus, there is a unique $k \in \mathbb{Z}$ such that $b = ka$.

Q.E.D.

Remark

The hypothesis $a \neq 0$ is exactly what the uniqueness step uses. If $a = 0$ and $b = 0$, then $0 \mid 0$, but *every* integer k satisfies $0 = k \cdot 0$.

Exercises

Use a proof by contradiction to show that two non-parallel lines have a unique intersection point.

4.5 If and Only If Proof

Recall

If and only if statement:

$$A \iff B \equiv (A \implies B) \wedge (B \implies A)$$

Example 4.5.1

Prove that for all $x, y \in \mathbb{R}$ with $x, y \geq 0$ that $\frac{x+y}{x} = \sqrt{xy}$ if and only if $x = y$.

Proof

Let x, y be arbitrary in $\mathbb{R}$ and non-negative.

Prove $(\iff)$ first. Assume $x = y$.

Then,

$$\frac{x+y}{2} = \frac{y+y}{2} = \frac{2y}{2} = y$$

and

$$\sqrt{xy} = \sqrt{yy} = \sqrt{y^2} = |y| = y.$$

since y is non-negative.

($\Rightarrow$) Assume $\frac{x+y}{2} = \sqrt{xy}$.

Then

$$x+y = 2\sqrt{xy} \implies 2x^2 + 2xy + y^2 = 4xy \implies x^2 - 2xy + y^2 = 0.$$

This implies

$$(x-y)^2 = 0.$$

Therefore,

$$x-y = 0 \implies x = y.$$

Q.E.D.

4.6 Sigma & Product Notation

Example 4.6.1

We have

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10$$

It is not too hard to write 10 numbers in a sum. What about

$$1 + 2 + \underbrace{\dots}_{\text{This is not rigorous}} + 1000$$

4.6.1 Summation Notation

Fill in rigorous definition.

$$1 + 2 + 3 + \dots + 100 = \sum_{i=1}^{100} i.$$

Example 4.6.2

$$1 + 8 + 27 + 64 + 125 = \sum_{i=1}^5 i^3$$

Properties of Summations

(i) Multiplying by a constant

$$c \sum_{i=a}^b t_i = \sum_{i=a}^b ct_i.$$

(ii) Addition or Subtraction of Summations

$$\sum_{j=1}^n x_i \pm \sum_{i=1}^n y_i = \sum_{i=1}^n (x_i \pm y_i)$$

(iii) Change bounds

$$\sum_{i=m}^n x_i = \sum_{i=m+r}^{n+r} x_{i-r}$$

for any $m, n, r \in \mathbb{Z}$.

(iv) Breaking Sum Apart

$$\sum_{i=1}^{n-1} i + \sum_n^k = \sum_{i=1}^k i.$$

Product Notation

$$\prod_{i=1}^n = 1 \cdot 2 \cdot 3 \cdots n = n!$$